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Paleomagnetism of Upper Carboniferous limestones from the Carnic Alps

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Abstract

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Limestones of Stephanian age belonging to the Carboniferous–Permian sedimentary sequence of the Auernig Group at Passo Pramollo (Carnic Alps) were investigated by a paleomagnetic study. These limestones represent the late-tectonic molassic stage of the Hercynian orogeny in the Paleozoic chain of the Carnic Alps. After a preliminary sampling at twenty sites, unstable magnetization was found to affect the remanence at eight sites. Of twelve sites studied in detail (about 180 specimens), 10 sites had a mean paleomagnetic direction of $D = 141.8°$ and $I = -8.1°$, and $\alpha_{95} = 17.8°$ after cleaning and corrections for bedding attitudes tilted by subsequent, mainly Alpine tectonics. The corresponding virtual mean pole position lies at 243.3° E, 36.2° N, coincident with data of comparable age from Morocco, and is located east of the African poles of Late Carboniferous to Permian age. In addition, the pole from the Pramollo limestones is also comparable with the younger Permian poles from the Southern Alps and from most Mediterranean crustal blocks, showing that the area was also rotated counterclockwise with respect to Europe. Paleogeographically, the area of the Carnic Alps already belonged to the Southern Alpine domain by Upper Carboniferous times, and was located in the Northern Hemisphere at that time.

Geology

In the Carnic Alps the Permian–Carboniferous sequence at Passo Pramollo (Nassfeld Schichten in the German literature: Heritsch, 1934; Auernig Group: Selli, 1963; see Fig. 1) near Pontebba has attracted the interest of geologists for many years because of two peculiar characteristics: its important and well preserved macrofloras and macrofaunas, and its little-disturbed, often subhorizontal attitudes. The Carnic Alps, the easternmost fragment of the Southern Alps s.s., underwent both the Hercynian and the Alpine orogenies.

The Permian–Carboniferous sequence represents the "molassic" sedimentation of the Hercynian cycle. During the Carboniferous the Auernig Group was affected by synsedimentary fault systems oriented N120° E and N15–40° E (Venturini, 1982, 1986). It was only during the Alpine orogeny that strong shortening took place, with the development of E–W-oriented folds and thrusts. These deformations are clearly recognized in the field; so it has been possible to select sites with well documented tectonic corrections.

In the Pramollo area, the environmental paleogeography, strongly controlled by tectonics, evolved from the initial proximal and intermediate alluvial fan deposition to later deltaic, paralic and shallow-water sedimentation. All erosional activity ceased in the Stephanian. Since then the earlier immature clastics have been followed by highly mature quartzose sediments supplied from the west

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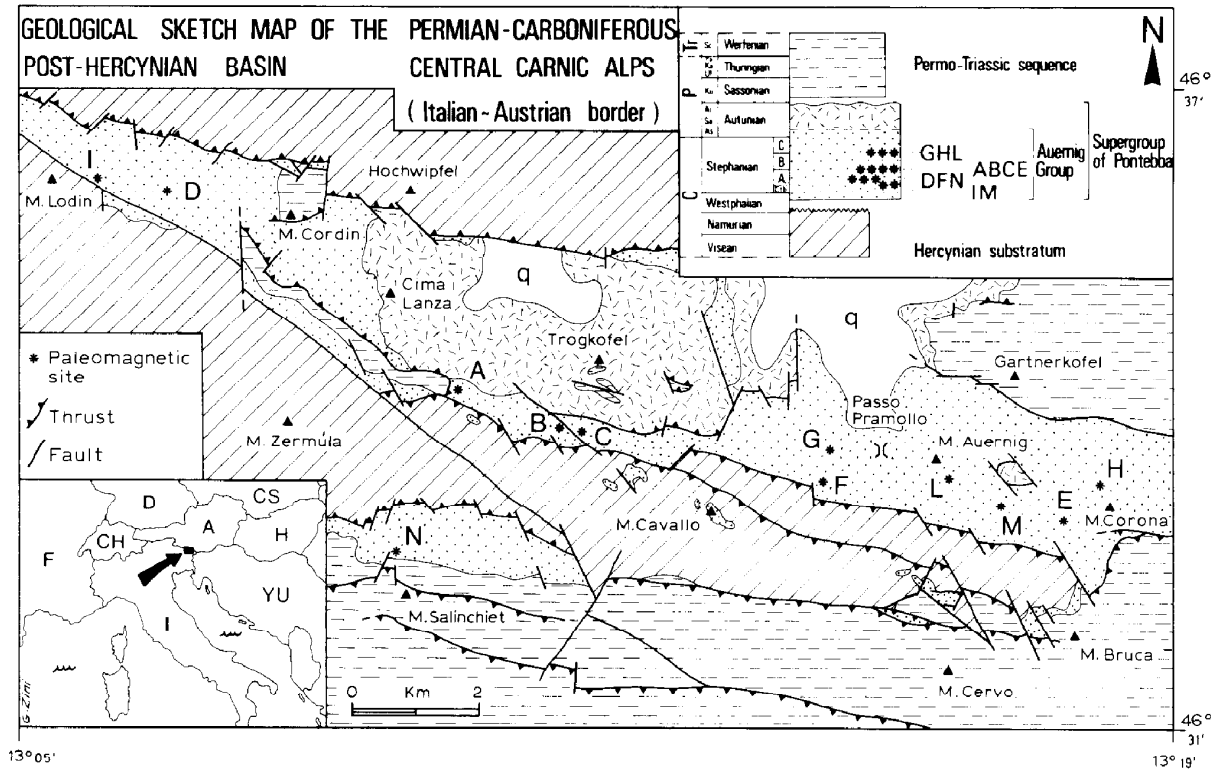


Fig. 1. Geological sketch-map showing the main structural elements of the Central Carnic Alps. In the stratigraphic column the sites sampled for paleomagnetic study are shown at their stratigraphic positions.

in flat deltas and shallow seas, alternating with shallow-water limestone beds.

The complete sequence, 100–1500 m thick in differentially sunken fault-blocks, is composed of four units (from top of bottom):

(d)	Trogkofel Limestones	(0–400 m)
(c)	Rattendorf Group	(0–460 m)
(b)	Auernig Group	(0–1100 m)
(a)	basal clastics	(0–200 m)

The whole sequence represents a Westphalian C – Lower Permian major transgressive cycle. It was followed later by a younger basal clastic horizon (Tarvisio Breccia, Sesto Conglomerate), and by Permian red beds (Val Gardena Sandstones) (Broglio Loriga et al., 1986).

Tests and measurements

For this study over 200 hand samples were collected from the limestone beds of the Auernig Group at 20 sites. After preliminary attempts on the different lithologies, the carbonate beds were selected as being most suitable for this study. They consist of well bedded, massive marine limestones organized in beds with thicknesses ranging from 2 to 15 m. Over 500 1-inch specimens were measured with a Jelinek JR-4 spinner magnetometer and demagnetized at the C.N.R. paleomagnetic laboratory in Bologna. However, due to random viscous components acquired during thermal demagnetization, a primary stable remanence could not be isolated in samples from several sites; the data suitable for interpretation were obtained from 119 samples at 12 sites.

The intensities of remanence range from 10^{-4} to 10^{-5} A/m, and are often highly variable between sites and even within a site. A few specimens showed negative values of magnetic susceptibility, certainly an unfavourable condition for subsequent study. NRM directions are highly dispersed (Fig. 2), attesting to the viscous and secondary components of magnetization. Three or more pilot specimens from each site have been

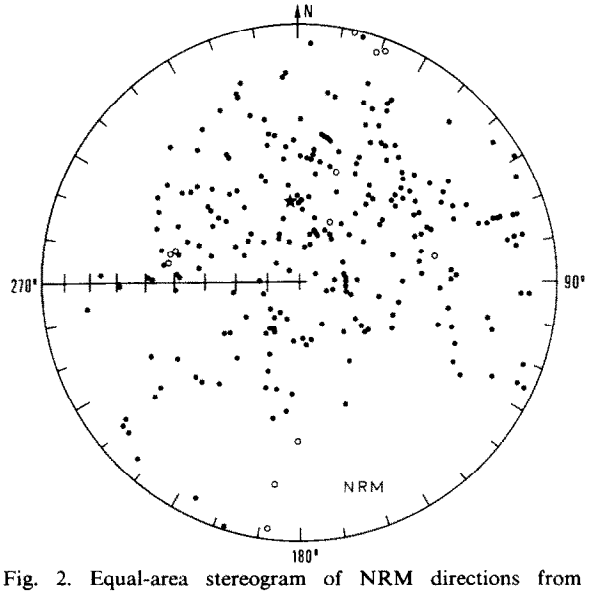


Fig. 2. Equal-area stereogram of NRM directions from Stephanian limestones at Pramollo. Full symbols—positive inclinations; open symbols—negative inclinations; star—present field direction.

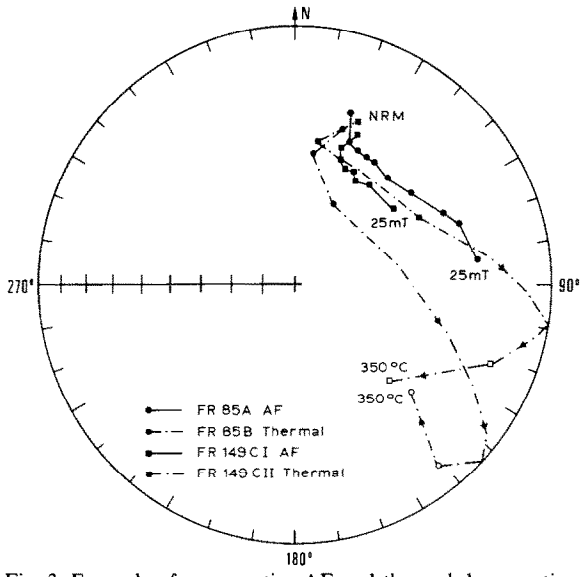


Fig. 3. Example of comparative AF and thermal demagnetizations. The endpoints of all curves correspond to low intensities of remanence, about 3% of NRM intensities.

subjected to progressive thermal demagnetization up to 500–600 °C and to alternating-field (AF) demagnetization in order to establish the best effective cleaning technique for obtaining stable characteristic remanent magnetizations (ChRM).

AF demagnetization was not successful in removing secondary magnetization components

TABLE 1

Paleomagnetic data and statistical parameters of Stephanian limestones from Carnic Alps, after thermal cleaning

Sites	N/n	Decl., D (°)	Incl., I (°)	Decl., D (°)	Incl., I (°)	R	k	α_{95}	Demagnetization (°C)
A	11/17	146.4	−26.3	152.5	−20.8	9.458	6.5	19.4	340–440
B	8/10	100.9	+22.2	100.8	+10.2	7.908	76.4	6.4	400–510
C	9/13	145.2	−37.2	150.2	−25.2	8.133	9.2	17.9	300–410
D	12/15	128.5	−4.6	127.2	+17.7	11.432	19.4	10.1	410–500
E	8/10	129.4	−15.9	133.6	−8.1	6.390	4.3	30.1	350–460
F	10/14	167.9	−17.7	166.2	+2.7	9.280	12.8	14.0	335–410
G	19/22	184.8	−31.4	155.4	−51.1	17.407	11.3	10.4	335–410
Selected	10/22	166.0	−35.7	134.8	−42.8	9.627	24.1	10.0	
H	7/ 9	345.8	+44.7	347.3	+62.7	6.851	40.2	9.6	400–440
I	16/19	118.7	−17.2	124.2	−1.8	15.493	29.6	6.9	350–410
L	11/14	266.9	−19.4	266.4	−5.1	10.227	6.2	19.0	340–480
M	10/18	158.8	−32.6	154.6	−19.1	9.241	11.9	14.6	250–300
N	14/24	174.7	−31.7	171.2	+10.1	12.252	7.4	15.6	400–470
Mean (excluding H and L)	10/12	142.9	−21.5	141.8	−8.1	8.916	8.3	17.8	

Pole position: lat. VGP = 36.2° N, long. VGP = 243.2° E

*—after tectonic corrections; N—number of measured specimens; n—number of specimens used in the calculation of the mean data. The last column gives the spectra of stable temperature fields.

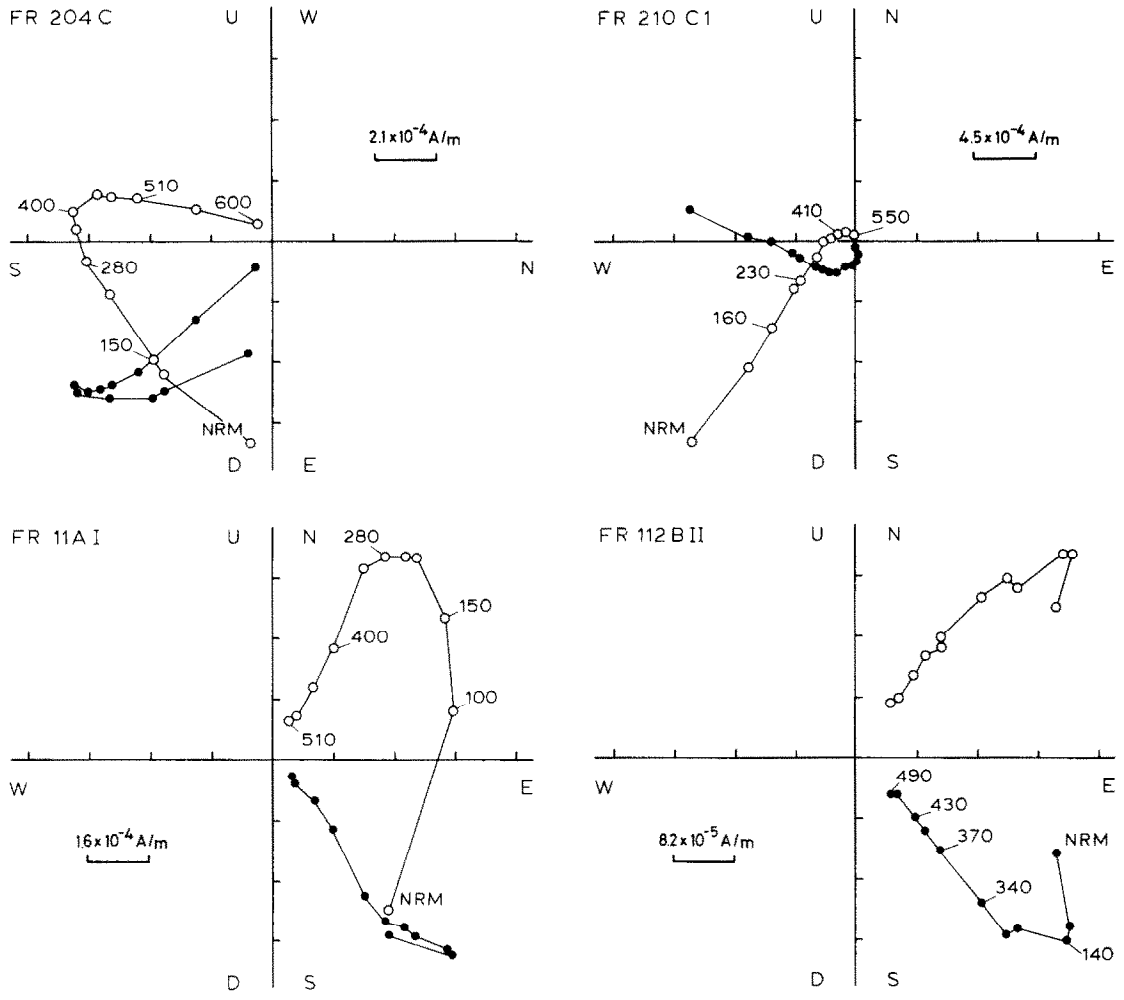


Fig. 4. Orthogonal projections of the vector endpoints of remanent magnetization during thermal cleaning. Dots—horizontal projections; circles—vertical projections. Steps are marked in °C.

completely (Fig. 3); in fact, specimens which were AF-demagnetized retained directions close to the NRM, whereas thermal cleaning of sister specimens from the same sample resulted in strong directional changes and complete removal of secondary components. Therefore thermal cleaning was preferred as the routine demagnetization technique. The most common types of demagnetization curves are plotted in Fig. 4. A shortcoming of thermal cleaning was the possible introduction of magnetization viscosity upon heating; above 500 °C it was impossible to obtain stable directions, so that further heating became useless. The conclusion of Lowrie and Heller (1982) that new fine-grained magnetite is formed in limestones during heating is accepted here. Different types of intensity decay curves which are found at the sampling sites are shown in Figs. 5a and b. Because of the strong variability of magnetic properties at a small scale the whole collection has been demagnetized in small steps; there is no common temperature spectrum within which all specimens reach stable directions of remanence (Table 1). Specimens which did not yield a stable endpoint during thermal cleaning or had stable, but random magnetizations, were discarded. Samples from eight sites were excluded from further study because NRM intensities had decreased to the magnetometer noise level at demagnetization temperatures of < 300 °C, or because the median destructive field (MDF) was below 7.5 mT. A few samples, with higher and very stable NRM (Fig. 4,

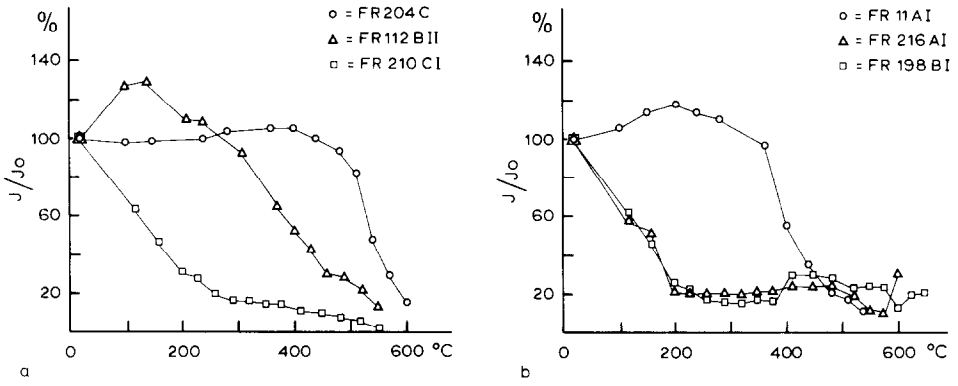


Fig. 5. Different types of NRM intensity thermal decay in the Stephanian limestones at Pramollo.

specimen FR210C1), exhibited higher blocking temperatures. After cleaning of these samples, they exhibited the same directions as specimens with weaker remanence and lower blocking temperatures. In conclusion, notwithstanding many cases of unfavourable conditions for measuring the remanent magnetization, stable and consistent ChRM has been measured in the Stephanian limestones at Pramollo. We have calculated the significant directions of remanence of each specimen by taking the mean of the directions obtained during cleaning steps removing stable magnetizations; these intervals are also reported in Table 1, together with the site mean directions and statistical parameters.

Paleomagnetic results

The mean directions and statistical parameters for all sites before and after tectonic corrections are shown in Table 1; Figs. 6a and b report the same data on stereographic projections. Values of α_{95} below 20° have been obtained for all sites except for site E. The mean direction of this site, despite the dispersion, is close to the overall mean.

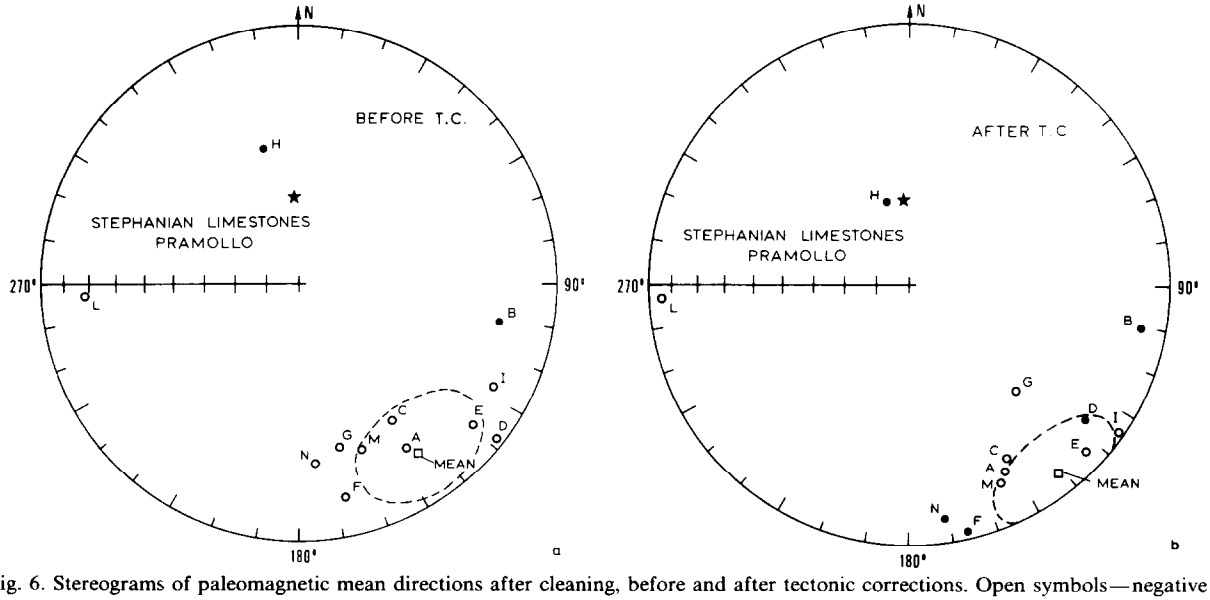


Fig. 6. Stereograms of paleomagnetic mean directions after cleaning, before and after tectonic corrections. Open symbols—negative inclinations; closed symbols—positive inclinations.

A few remarks on individual sites are necessary. Site G can be divided into two groups. One group of samples carries well aligned ChRM directions, whereas the second group contains more dispersed directions; both site means have been calculated with and without the second group (Table 1). The directions of sites H and L are widely separated from the cluster of directions of all other sites; we have interpreted these data as directions of remagnetization; however, the direction of site L, which is roughly opposite to all the other directions, could be interpreted as being of normal polarity; the directions of site H are close to Tertiary directions in the Alps, and are found in the NRM of several samples. Both sites, as well as site G, were sampled from a thick limestone bed, the lowest bed of the Auernig Formation, the magnetization of which is less consistent and, therefore, is less reliable than that of the other Stephanian limestones. The three sites D, F and N with positive mean inclinations after tilt correction also belong to a single thick limestone bed (Fig. 1), the second bed of the Lower Stephanian Pizzul Formation.

Sedimentary perturbances such as the original dip of the bedding, synsedimentary tectonic tilting or deposition in strong water currents may have distorted the remanence inclinations at sites D, F and N. Local and regional tectonics can adequately explain the anomalous paleomagnetic direction found at site B, which is displaced to the east ($D = 100.8°$, $I = 10.2°$). The directions obtained at site L ($D = 266.4°$, $I = -5.1°$) are similar to those at site B and coincide with directions found by Mauritsch and Becke (1983) in a section of the younger (Permian) Val Gardena Formation near Paularo ($D = 98.7°$, $I = -4.0°$; $D = 274.9°$, $I = 6.1°$). The counterclockwise displacement of these directions may be explained by local movements along Alpine transcurrent faults oriented N145° E.

The mean paleomagnetic directions from all other sites are located in the southeastern quadrant with low negative inclinations, which is in agreement with the reversed field polarity of the Kiaman superchron. Corrections for dip do not improve the clustering of directions (Table 1). Although the fold test is inconclusive, the low inclinations provide evidence that the cleaned directions can be primary or at least pre-Triassic. The folding age is Alpine (post-Cretaceous) and a possible post-folding remagnetization should provide steeper inclination values.

The overall mean direction calculated from ten sites of the Stephanian limestones at Pramollo is: $D = 141.8°$, $I = -8.1°$ and $\alpha_{95} = 17.8°$, and the calculated Stephanian paleopole position is: VGP lat. = 36.2° N, VGP long. = 343.3° E.

Discussion

Permian–Carboniferous paleomagnetic data from the Southern Alps are abundant. They have been derived primarily from the widespread ignimbritic flows and date back to the early years of paleomagnetism. These data are consistent all over the area, from the Southern Tessin Alps to the Dolomites and to the Vicentine Prealps (Dietzel, 1960; Van Hilten, 1960; De Boer, 1963; Zijderveld and De Jong, 1969; Manzoni, 1970; Zijderveld et al., 1970; Heiniger, 1979), as revisited and reported in the review paper by VandenBerg and Zijderveld (1982).

Previous to our study a very early paleomagnetic study was made in this area by Guicherit (1964). His results ($D = 24°$, $I = +40.5°$) were not consistent with all the other results from the Southern Alps. Moreover, his results implied that the Carnic Alps belonged to the European Plate and not to Africa, in contrast to all other results from the Southern Alps. This anomaly is not supported by geological data. The results published by Guicherit came from the same area which we have studied, but from a different lithology (sandstones of the Auernig Group). The different final results are possibly due to different cleaning techniques, because the directions published by Guicherit were derived from only nine samples cleaned by ineffective AF demagnetization.

The Upper Carboniferous directions from Pramollo are close to the known Permian data from the Southern Alps (Fig. 7); therefore we infer that the Carnic Alps have belonged to the domain of the Southern Alps since Late Carboniferous times. The paleolatitude of 4° N obtained from the Pramollo data is lower than the known

Permian paleolatitudes from the Southern Alps; this is in agreement with the subequatorial position and migration toward the north of the region during Late Carboniferous and Permian times (Briden et al., 1974; Scotese et al., 1979; Krs, 1982; Scotese, 1984). In turn, the Southern Alps were stable with respect to Africa. The pole position derived from the Pramollo limestones is close to the Late Carboniferous paleopole of the African continent, and even closer to the paleopole positions of Stephanian age from Morocco (Daly and Pozzi, 1976; Martin et al., 1978; Westphal et al., 1979; and see Fig. 8). Lowrie (1985) has shown the exact coincidence of the Mesozoic data from the Southern Alps with the apparent polar wander path (APWP) of Africa derived from the North American path. Zijderveld and Van der Voo (1973) proposed that the Southern Alps constituted a northern extension of the African Plate. It should be kept in mind that Pangea (model A2) had just assembled in the Late Carboniferous (Van der Voo et al., 1984; Kent and Keppie, in press), and that the movements between Gondwana and Laurussia are thought to have terminated in Stephanian times (Van der Voo, 1982).

Our data sustain the hypothesis of a Southern Alpine domain, including the Carnic Alps, lying within the African Plate since the Late Paleozoic; thus its poles are also representative of the African APWP. The coincidence of the South Alpine poles with the Moroccan poles (Fig. 8) provides evidence that the latter are also representative of the African continent. This view is reinforced by the coincidence of the Moroccan poles with synchronous South American poles prior to the opening of the Atlantic.

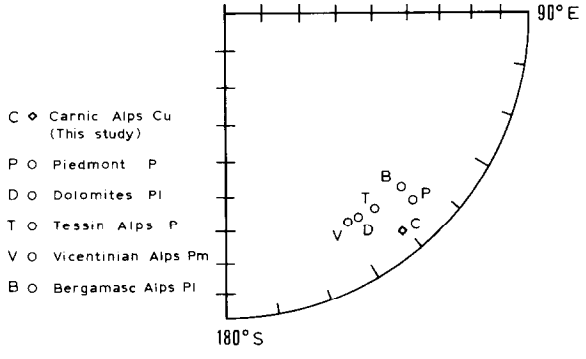


Fig. 7. Upper Paleozoic paleomagnetic directions for the Southern Alps. Carnic Alps—this study; Piedmont—Heiniger (1979); Dolomites—Van Hilten (1960), Zijderveld et al. (1970); Tessin Alps—Van Hilten and Zijderveld (1966), Heiniger (1979); Vicentinian Alps—De Boer (1963); Bergamasc Alps—Zijderveld and De Jong (1969).

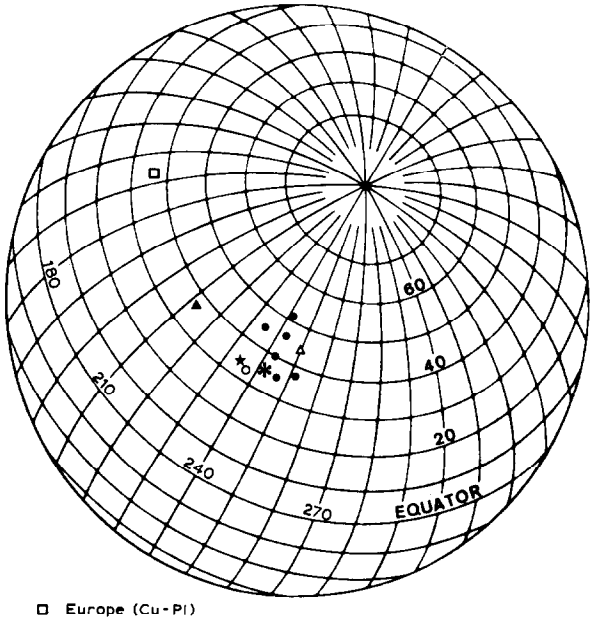


Fig. 8. Equal-area projection of Upper Paleozoic paleomagnetic poles from Europe, Africa, Sardinia and the Alps. African pole transferred from North American APWP after Van der Voo and French (1974) (Pangea A2).

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